GMC LANKA CREATIONS

Test Case Register

SENG 34213 – System Development Project

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A report submitted in partial fulfillment of the requirements for the degree of Bachelor of Science Honors in Software Engineering (B. Sc.SE)

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2026

Version history

Version No.	Date	Author	Summary of Changes
01	2026/10/24	Team	Register generated from the test run of 24 October 2026

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CHAPTER 1 INTRODUCTION

This register documents every automated test case of the GMC Lanka Creations system together with the requirement it verifies and its actual result. It satisfies guideline Section 6.3.3 and deliverable 7 of Section 10.1. The results come from the test execution of 24 October 2026 on the release code: backend unit tests (Maven Surefire), backend API integration tests (Maven Failsafe) and frontend tests (Vitest).

Table 1.1: Execution summary

Suite	Location	Test cases	Passed	Failed	Skipped
Backend unit	backend/src/test/java	218	218	0	0
Backend integration	tests/backend/src/test/java	253	253	0	0
Frontend unit + integration	frontend/src, tests/frontend	37	37	0	0
Total		**508**	**508**	**0**	**0**

Test case identifiers follow the pattern TC-<area>-<number> for API tests, UT-<area>-<number> for backend unit tests and FE-<number> for frontend tests. Names follow the convention method_condition_expectedResult; the condition is the precondition or input and the last part is the expected output.

CHAPTER 2 DETAILED TEST CASES (PRIORITY 1)

The following functional test cases cover the critical paths of the system and are documented in the format of guideline Section 6.3.3.

TC-FR-001-01 – Successful registration

Field	Content
Test Case ID	TC-FR-001-01
Test Case Name	Successful registration
Related Requirement	FR001
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/AuthControllerIT.java
Test Method	register_withValidDetails_returns201AndCustomerToken
Type	Integration
Priority	P1 – Must Pass
Preconditions	No account exists for the e-mail address
Input	POST /api/auth/register with name, valid e-mail and 8+ character password
Expected Output	HTTP 201; access token returned; account created with role CUSTOMER
Actual Output	As expected
Status	Pass

TC-FR-001-02 – Duplicate e-mail rejected

Field	Content
Test Case ID	TC-FR-001-02
Test Case Name	Duplicate e-mail rejected
Related Requirement	FR001
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/AuthControllerIT.java
Test Method	register_withDuplicateEmail_isRejected
Type	Integration
Priority	P1 – Must Pass
Preconditions	An account already exists for the e-mail
Input	POST /api/auth/register with the same e-mail
Expected Output	HTTP 409 DUPLICATE_EMAIL; no second account
Actual Output	As expected
Status	Pass

TC-FR-003-01 – Login sets HttpOnly refresh cookie

Field	Content
Test Case ID	TC-FR-003-01
Test Case Name	Login sets HttpOnly refresh cookie
Related Requirement	FR003
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/AuthControllerIT.java
Test Method	login_withCorrectPassword_returnsTokenAndHttpOnlyRefreshCookie
Type	Integration
Priority	P1 – Must Pass

Field	Content
Preconditions	Active customer account
Input	POST /api/auth/login with correct credentials
Expected Output	HTTP 200; access token in body; refreshToken cookie with HttpOnly
Actual Output	As expected
Status	Pass

TC-FR-003-02 – Brute-force protection

Field	Content
Test Case ID	TC-FR-003-02
Test Case Name	Brute-force protection
Related Requirement	FR003, NFR011
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/AuthControllerIT.java
Test Method	login_afterTenFailedAttempts_returns429
Type	Integration
Priority	P1 – Must Pass
Preconditions	Ten failed logins for the same IP and e-mail within 15 minutes
Input	Eleventh POST /api/auth/login
Expected Output	HTTP 429 TOO_MANY_REQUESTS
Actual Output	As expected
Status	Pass

TC-FR-003-03 – Password reset does not reveal accounts

Field	Content
Test Case ID	TC-FR-003-03
Test Case Name	Password reset does not reveal accounts
Related Requirement	FR003
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/AuthControllerIT.java
Test Method	forgotPassword_forUnknownEmail_givesSameReplyAndSendsNothing
Type	Integration
Priority	P1 – Must Pass
Preconditions	E-mail address not registered
Input	POST /api/auth/forgot-password
Expected Output	Same generic reply as for a known e-mail; no e-mail sent
Actual Output	As expected
Status	Pass

TC-FR-002-01 – Customer cannot reach admin endpoints

Field	Content
Test Case ID	TC-FR-002-01
Test Case Name	Customer cannot reach admin endpoints
Related Requirement	FR002, NFR011
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/AdminControllerIT.java
Test Method	adminEndpoints_withCustomerToken_return403
Type	Integration

Field	Content
Priority	P1 – Must Pass
Preconditions	Logged in as CUSTOMER
Input	Requests to 10 /api/admin endpoints
Expected Output	HTTP 403 FORBIDDEN for every endpoint
Actual Output	As expected
Status	Pass

TC-FR-005-01 – Customer cannot create products

Field	Content
Test Case ID	TC-FR-005-01
Test Case Name	Customer cannot create products
Related Requirement	FR005, NFR011
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/ProductControllerIT.java
Test Method	create_asCustomer_returns403
Type	Integration
Priority	P1 – Must Pass
Preconditions	Logged in as CUSTOMER
Input	POST /api/products
Expected Output	HTTP 403; no product created
Actual Output	As expected
Status	Pass

TC-FR-012-01 – Cash-on-delivery order

Field	Content
Test Case ID	TC-FR-012-01
Test Case Name	Cash-on-delivery order
Related Requirement	FR012
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java
Test Method	placeOrder_cashOnDelivery_createsOrderReservesStockAndEmptiesCart
Type	Integration
Priority	P1 – Must Pass
Preconditions	Cart with in-stock items
Input	POST /api/orders with shipping details and CASH_ON_DELIVERY
Expected Output	HTTP 201; order PENDING; stock reduced; cart emptied
Actual Output	As expected
Status	Pass

TC-FR-012-02 – Order rejected when stock ran out

Field	Content
Test Case ID	TC-FR-012-02
Test Case Name	Order rejected when stock ran out
Related Requirement	FR012
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java
Test Method	placeOrder_whenStockRanOutAfterAddingToCart_returns422AndKeepsCart

Field	Content
Type	Integration
Priority	P1 – Must Pass
Preconditions	Stock reduced below the cart quantity after adding to cart
Input	POST /api/orders
Expected Output	HTTP 422; no order; cart kept
Actual Output	As expected
Status	Pass

TC-FR-012-03 – Order with empty cart rejected

Field	Content
Test Case ID	TC-FR-012-03
Test Case Name	Order with empty cart rejected
Related Requirement	FR012
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java
Test Method	placeOrder_withEmptyCart_returns422
Type	Integration
Priority	P1 – Must Pass
Preconditions	Customer's cart is empty
Input	POST /api/orders
Expected Output	HTTP 422 BUSINESS_RULE_VIOLATION
Actual Output	As expected
Status	Pass

TC-FR-014-01 – Other customers' orders hidden

Field	Content
Test Case ID	TC-FR-014-01
Test Case Name	Other customers' orders hidden
Related Requirement	FR014, NFR011
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java
Test Method	getOrder_anotherCustomersOrder_returns404
Type	Integration
Priority	P1 – Must Pass
Preconditions	Order belongs to another customer
Input	GET /api/orders/{id}
Expected Output	HTTP 404; order details not disclosed
Actual Output	As expected
Status	Pass

TC-FR-018-01 – Card payment confirmed by Stripe

Field	Content
Test Case ID	TC-FR-018-01
Test Case Name	Card payment confirmed by Stripe
Related Requirement	FR018
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java

Field	Content
Test Method	confirmPayment_whenStripeReportsSuccess_marksOrderPaid
Type	Integration
Priority	P1 – Must Pass
Preconditions	Unpaid card order with a PaymentIntent
Input	POST /api/orders/{id}/payment/confirm; Stripe reports succeeded
Expected Output	Order payment status COMPLETED
Actual Output	As expected
Status	Pass

TC-FR-018-02 – Mismatched payment refunded

Field	Content
Test Case ID	TC-FR-018-02
Test Case Name	Mismatched payment refunded
Related Requirement	FR018
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java
Test Method	confirmPayment_whenAmountDoesNotMatch_refundsInsteadOfMarkingPaid
Type	Integration
Priority	P1 – Must Pass
Preconditions	Stripe reports a charge whose amount differs from the order total
Input	POST /api/orders/{id}/payment/confirm
Expected Output	Payment refunded; order not marked paid
Actual Output	As expected
Status	Pass

TC-FR-018-03 – Retried payment reuses intent

Field	Content
Test Case ID	TC-FR-018-03
Test Case Name	Retried payment reuses intent
Related Requirement	FR018
Related Test File	tests/backend/src/test/java/com/gmc/backend/it/OrderControllerIT.java
Test Method	startPayment_retried_reusesOpenPaymentInsteadOfCreatingAnother
Type	Integration
Priority	P1 – Must Pass
Preconditions	Payment already started for the order
Input	POST /api/orders/{id}/payment again
Expected Output	Same PaymentIntent returned; no second charge possible
Actual Output	As expected
Status	Pass

CHAPTER 3 BACKEND API INTEGRATION TEST CASES

3.1 AuthControllerIT (/api/auth/*) – FR001, FR003

ID	Operation	Condition / input	Expected output	Status
TC-AUTH-01	Refresh	With Valid Cookie	Returns New Access Token	Pass
TC-AUTH-02	Protected Endpoint	Without Token	Is Denied	Pass
TC-AUTH-03	Reset Password	With Emailed Token	Changes Password Once	Pass
TC-AUTH-04	Login	With Missing Fields	Returns400	Pass
TC-AUTH-05	Login	With Wrong Password	Returns401	Pass
TC-AUTH-06	Refresh	With Unknown Token	Returns401	Pass
TC-AUTH-07	Reset Password	With Invalid Token	Is Rejected	Pass
TC-AUTH-08	Refresh	Without Cookie	Returns401	Pass
TC-AUTH-09	Logout	Without Cookie	Returns204	Pass
TC-AUTH-10	Forgot Password	For Unknown Email	Gives Same Reply And Sends Nothing	Pass
TC-AUTH-11	Register	With Duplicate Email	Is Rejected	Pass
TC-AUTH-12	Login	After Ten Failed Attempts	Returns429	Pass
TC-AUTH-13	Forgot Password	For Known Email	Sends Reset Link With Generic Reply	Pass
TC-AUTH-14	Register	With Short Password And Bad Email	Returns400	Pass
TC-AUTH-15	Register	With Valid Details	Returns201 And Customer Token	Pass
TC-AUTH-16	Logout	—	Invalidates Refresh Token	Pass
TC-AUTH-17	Protected Endpoint	With Wrong Role	Returns403	Pass
TC-AUTH-18	Login	With Correct Password	Returns Token And Http Only Refresh Cookie	Pass

3.2 UserControllerIT (/api/users/me/*) – FR003, FR027

ID	Operation	Condition / input	Expected output	Status
TC-USER-01	Mark Read	Own Notification	Decrements Unread Count	Pass
TC-USER-02	Change Password	With Wrong Current	Returns422 And Keeps Password	Pass
TC-USER-03	Mark All Read	—	Clears Unread Count	Pass
TC-USER-04	Get Unread Count	—	Counts Unread Notifications	Pass
TC-USER-05	Update Profile	With Blank Name	Keeps Existing Name	Pass
TC-USER-06	Get Profile	—	Returns Callers Details Without Password	Pass
TC-USER-07	Upload Avatar	With Disallowed Type	Returns400	Pass
TC-USER-08	Change Password	With Correct Current	Revokes Old Sessions And Starts New One	Pass
TC-USER-09	Upload Avatar	With Empty File	Returns400	Pass
TC-USER-10	Mark Read	Someone Elses Notification	Leaves It Unread	Pass
TC-USER-11	Get Profile	Anonymous	Returns403	Pass
TC-USER-12	Get Notifications	—	Returns Only Callers Notifications	Pass
TC-USER-13	Update Profile	With Too Long Phone	Returns400	Pass
TC-USER-14	Remove Avatar	—	Clears Profile Image Url	Pass
TC-USER-15	Get Profile	With Tampered Token	Returns401	Pass
TC-USER-16	Remove Avatar	Anonymous	Returns403	Pass
TC-USER-17	Upload Avatar	With Png	Sets Profile Image Url	Pass
TC-USER-18	Change Password	With Short New Password	Returns400	Pass
TC-USER-19	Notifications	Anonymous	Returns403	Pass
TC-USER-20	Get Profile	—	Works For Every Role	Pass

ID	Operation	Condition / input	Expected output	Status
TC-USER-21	Update Profile	—	Changes Provided Fields Only	Pass

3.3 ProductControllerIT (/api/products/*) – FR005–FR008

ID	Operation	Condition / input	Expected output	Status
TC-PROD-01	Get All	By Unknown Category	Returns404	Pass
TC-PROD-02	Update	Unknown Product	Returns404	Pass
TC-PROD-03	Delete	As Admin	Archives Product But Keeps It	Pass
TC-PROD-04	Upload Images	As Customer	Returns403	Pass
TC-PROD-05	Get All	By Material	Filters Case Insensitively	Pass
TC-PROD-06	Set Active	As Admin	Restores Archived Product	Pass
TC-PROD-07	Create	Anonymous	Returns403	Pass
TC-PROD-08	Update	As Technician	Returns403	Pass
TC-PROD-09	Create	With Unknown Category	Returns404	Pass
TC-PROD-10	Upload Images	As Admin	Stores Images And Returns Urls	Pass
TC-PROD-11	Create	As Customer	Returns403	Pass
TC-PROD-12	Update	As Admin	Changes Fields And Category	Pass
TC-PROD-13	Update	With Negative Stock	Returns400	Pass
TC-PROD-14	Get By Id	Unknown Product	Returns404	Pass
TC-PROD-15	Create	As Admin	Returns201 With Cover Image	Pass
TC-PROD-16	Get All	By Search	Matches Name Ignoring Case	Pass
TC-PROD-17	Delete	As Customer	Returns403	Pass
TC-PROD-18	Get By Id	Non Numeric Id	Returns400	Pass
TC-PROD-19	Get Reviews	With Visible And Hidden Reviews	Returns Only Visible With Average	Pass
TC-PROD-20	Delete	Unknown Product	Returns404	Pass
TC-PROD-21	Get All	Anonymous	Returns Only Active Products	Pass
TC-PROD-22	Set Active	As Customer	Returns403	Pass
TC-PROD-23	Get All	By Category	Returns Active Products In That Category	Pass
TC-PROD-24	Create	With Missing Name And Zero Price	Returns400	Pass
TC-PROD-25	Get Reviews	Product Without Reviews	Returns Zero Average	Pass
TC-PROD-26	Set Active	False	Hides Product From Shop	Pass
TC-PROD-27	Get By Id	Existing Product	Returns Details	Pass
TC-PROD-28	Get All Categories	Anonymous	Lists Categories	Pass
TC-PROD-29	Set Active	Without Active Param	Returns400	Pass
TC-PROD-30	Get Reviews	Unknown Product	Returns404	Pass
TC-PROD-31	Upload Images	With Disallowed Extension	Returns400	Pass

3.4 CartControllerIT (/api/cart/*) – FR012

ID	Operation	Condition / input	Expected output	Status
TC-CART-01	Get Cart	For New Customer	Returns Empty Cart	Pass
TC-CART-02	Update Quantity	Without Quantity Param	Returns400	Pass
TC-CART-03	Remove Item	Own Item	Removes It	Pass
TC-CART-04	Update Quantity	Above Stock	Returns422	Pass
TC-CART-05	Update Quantity	On Another Customers Item	Is Rejected And Item Unchanged	Pass
TC-CART-06	Clear Cart	Without Token	Is Denied	Pass

ID	Operation	Condition / input	Expected output	Status
TC-CART-07	Clear Cart	—	Empties Cart	Pass
TC-CART-08	Remove Item	Unknown Item	Returns404	Pass
TC-CART-09	Add Item	Unknown Product	Returns404	Pass
TC-CART-10	Update Quantity	Unknown Item	Returns404	Pass
TC-CART-11	Add Item	With Stock	Adds Item And Totals	Pass
TC-CART-12	Add Item	Same Product Twice	Merges Quantity	Pass
TC-CART-13	Update Quantity	Within Stock	Updates Item	Pass
TC-CART-14	Get Cart	As Admin	Returns403	Pass
TC-CART-15	Add Item	With Zero Quantity Or Missing Product	Returns400	Pass
TC-CART-16	Add Item	More Than In Stock	Returns422	Pass
TC-CART-17	Add Item	Inactive Product	Returns422	Pass
TC-CART-18	Remove Item	Another Customers Item	Returns422	Pass
TC-CART-19	Update Quantity	To Zero	Removes Item	Pass

3.5 OrderControllerIT (/api/orders/*) – FR012, FR014, FR017, FR018

ID	Operation	Condition / input	Expected output	Status
TC-ORD-01	Get Order	Own Order	Returns Details	Pass
TC-ORD-02	Get Order	Another Customers Order	Returns404	Pass
TC-ORD-03	Start Payment	When Already Paid At Stripe	Marks Order Paid	Pass
TC-ORD-04	Checkout Summary	As Technician	Returns403	Pass
TC-ORD-05	Place Order	With Empty Cart	Returns422	Pass
TC-ORD-06	Confirm Payment	When Customer Has Not Paid Yet	Leaves Order Pending	Pass
TC-ORD-07	Start Payment	Retried	Reuses Open Payment Instead Of Creating Another	Pass
TC-ORD-08	Start Payment	For Cash On Delivery Order	Returns422	Pass
TC-ORD-09	Confirm Payment	When Amount Does Not Match	Refunds Instead Of Marking Paid	Pass
TC-ORD-10	Place Order	When Stock Ran Out After Adding To Cart	Returns422 And Keeps Cart	Pass
TC-ORD-11	Get My Orders	—	Returns Only The Customers Own Orders	Pass
TC-ORD-12	Confirm Payment	When Stripe Reports Success	Marks Order Paid	Pass
TC-ORD-13	Get Order	Unknown Or Non Numeric Id	Returns404 Or400	Pass
TC-ORD-14	Checkout Summary	With Cart Items	Returns Price Breakdown	Pass
TC-ORD-15	Place Order	By Card When Stripe Not Configured	Returns422	Pass
TC-ORD-16	Start Payment	For Another Customers Order	Returns404	Pass
TC-ORD-17	Place Order	With Invalid Phone Or Missing Payment Method	Returns400	Pass
TC-ORD-18	Place Order	Without Token	Is Denied	Pass
TC-ORD-19	Start Payment	For Unpaid Card Order	Returns Card Form Details	Pass
TC-ORD-20	Checkout Summary	With Stripe Configured	Offers Card Payment	Pass
TC-ORD-21	Confirm Payment	For Another Customers Order	Returns404	Pass
TC-ORD-22	Place Order	Cash On Delivery	Creates Order Reserves Stock And Empties Cart	Pass
TC-ORD-23	Place Order	By Card	Creates Unpaid Order Without Charging	Pass

3.6 PaymentControllerIT (/api/payments/stripe/webhook) – FR018

ID	Operation	Condition / input	Expected output	Status
TC-PAY-01	Webhook	Without Signature Header	Returns400	Pass

ID	Operation	Condition / input	Expected output	Status
TC-PAY-02	Webhook	With Invalid Signature	Returns400	Pass
TC-PAY-03	Webhook	Payment Canceled	Marks Order Payment Failed	Pass
TC-PAY-04	Other Payment Endpoints	—	Require Customer Login	Pass
TC-PAY-05	Webhook	Payment Succeeded	Marks Matching Order Paid Without Login	Pass
TC-PAY-06	Webhook	When Webhook Secret Not Configured	Returns503	Pass
TC-PAY-07	Webhook	For Unrelated Event Or Unknown Payment	Returns200 And Changes Nothing	Pass

3.7 AppointmentControllerIT (/api/appointments/*) – FR023, FR024

ID	Operation	Condition / input	Expected output	Status
TC-APT-01	Appointments	As Admin Or Anonymous	Returns403	Pass
TC-APT-02	Cancel	Completed Appointment	Returns422	Pass
TC-APT-03	Reschedule	Someone Elses Appointment	Returns404	Pass
TC-APT-04	Cancel	Unknown Appointment	Returns404	Pass
TC-APT-05	Reschedule	Cancelled Appointment	Returns422	Pass
TC-APT-06	Cancel	Own Pending Appointment	Returns Cancelled	Pass
TC-APT-07	Get My Appointments	—	Returns Only Callers Appointments	Pass
TC-APT-08	Upload Images	With More Than Six Files	Returns422	Pass
TC-APT-09	Book	With Foreign Image Url	Returns422	Pass
TC-APT-10	Cancel	Someone Elses Appointment	Returns404 And Leaves It Unchanged	Pass
TC-APT-11	Book	With Uploaded Images And Category	Returns201 Pending	Pass
TC-APT-12	Reschedule	Own Approved Appointment	Moves Back To Pending	Pass
TC-APT-13	Upload Images	With Png Files	Returns201 And Appointment Urls	Pass
TC-APT-14	Reschedule	With Past Date	Returns400	Pass
TC-APT-15	Book	With Past Date Or Blank Message	Returns400	Pass
TC-APT-16	Book	With Unknown Category	Returns422	Pass
TC-APT-17	Upload Images	With Disallowed Extension	Returns400	Pass

3.8 FeedbackControllerIT (/api/feedback/*) – FR029

ID	Operation	Condition / input	Expected output	Status
TC-FBK-01	Submit	As Admin	Returns403	Pass
TC-FBK-02	Update	With Rating Zero	Returns400	Pass
TC-FBK-03	Update	Own Review	Changes Rating And Marks Edited	Pass
TC-FBK-04	Get My Feedback	As Technician	Returns403	Pass
TC-FBK-05	Submit	For Product Not In Order	Returns422	Pass
TC-FBK-06	Get My Feedback	—	Returns Only Callers Reviews	Pass
TC-FBK-07	Update	Unchanged Review	Keeps Updated At Empty	Pass
TC-FBK-08	Submit	For Someone Elses Order	Returns422	Pass
TC-FBK-09	Submit	Twice For Same Product	Returns422	Pass
TC-FBK-10	Submit	For Undelivered Order	Returns422	Pass
TC-FBK-11	Submit	For Unknown Order	Returns404	Pass
TC-FBK-12	Update	Someone Elses Review	Returns422	Pass
TC-FBK-13	Submit	Low Rating	Notifies Admins	Pass
TC-FBK-14	Submit	With Rating Out Of Range And	Returns400	Pass

ID	Operation	Condition / input	Expected output	Status
		Blank Comment		
TC-FBK-15	Update	Unknown Review	Returns404	Pass
TC-FBK-16	Submit	For Delivered Product	Returns201	Pass
TC-FBK-17	Submit	Anonymous	Returns403	Pass

3.9 ComplaintControllerIT (/api/complaints/*) – FR028

ID	Operation	Condition / input	Expected output	Status
TC-CMP-01	Submit	With Product Not In Order	Returns422	Pass
TC-CMP-02	Complaints	As Technician	Returns403	Pass
TC-CMP-03	Get My Complaints	—	Returns Only Callers Complaints	Pass
TC-CMP-04	Submit	About Someone Elses Order	Returns422	Pass
TC-CMP-05	Submit	About Product In Own Order	Links Order And Product	Pass
TC-CMP-06	Submit	Without Order	Returns201 Open	Pass
TC-CMP-07	Submit	With Blank Description	Returns400	Pass

3.10 ContactControllerIT (/api/contact) – FR028

ID	Operation	Condition / input	Expected output	Status
TC-CON-01	Send	Sixth Message From Same Ip	Returns429	Pass
TC-CON-02	Send	Without Phone	Is Accepted	Pass
TC-CON-03	Send	Anonymous	Notifies Admins In App And By Email	Pass
TC-CON-04	Send	With Invalid Email Short Message And Bad Phone	Returns400	Pass
TC-CON-05	Send	Fourth Message From Same Email	Returns429	Pass

3.11 CustomDesignControllerIT (/api/custom-designs/*) – FR009

ID	Operation	Condition / input	Expected output	Status
TC-CDS-01	Submit	Without File	Returns201 With No File Path	Pass
TC-CDS-02	Submit	With Svg File	Returns400	Pass
TC-CDS-03	Submit	With Pdf File	Returns201 Pending Review And Stores File	Pass
TC-CDS-04	Submit	Without Design Name	Returns400	Pass
TC-CDS-05	Get My Designs	—	Returns Only Callers Designs	Pass
TC-CDS-06	Custom Designs	As Technician	Returns403	Pass

3.12 TechnicianControllerIT (/api/technician/*) – FR020

ID	Operation	Condition / input	Expected output	Status
TC-TEC-01	Update Task Status	With Unknown Status	Returns422	Pass
TC-TEC-02	Update Task Status	Without Status Param	Returns400	Pass
TC-TEC-03	Update Task Status	On Unknown Task	Returns404	Pass
TC-TEC-04	Update Task Status	To In Progress	Returns200	Pass
TC-TEC-05	Update Task Status	Lowercase Completed	Is Normalised And Saved	Pass
TC-TEC-06	Technician Endpoints	As Customer	Returns403	Pass
TC-TEC-07	Get My Tasks	—	Returns Only Tasks Assigned	Pass

ID	Operation	Condition / input	Expected output	Status
			To Caller	
TC-TEC-08	Update Task Status	On Completed Task	Returns422	Pass
TC-TEC-09	Update Task Status	On Colleagues Task	Returns422 And Leaves It Unchanged	Pass

3.13 AdminControllerIT (/api/admin/*) – FR002, FR004, FR010, FR015, FR016, FR020, FR026, FR030

ID	Operation	Condition / input	Expected output	Status
TC-ADM-01	Update Complaint Status	With Unknown Status	Returns422	Pass
TC-ADM-02	Update Order Status	Skipping Steps	Returns422	Pass
TC-ADM-03	Create Category	With New Name	Returns201	Pass
TC-ADM-04	Get All Appointments	Without Filter	Lists Appointments	Pass
TC-ADM-05	Activate User	When User Missing	Returns404	Pass
TC-ADM-06	Deactivate User	For Another User	Marks Inactive	Pass
TC-ADM-07	Get All Orders	Without Filter	Lists Orders	Pass
TC-ADM-08	Delete Category	When Used By Product	Returns422	Pass
TC-ADM-09	Assign Maintenance	To Technician	Returns201 And Lists It	Pass
TC-ADM-10	Update Order Status	Without Status	Returns400	Pass
TC-ADM-11	Get All Products For Admin	—	Includes Archived Products	Pass
TC-ADM-12	Delete Category	When Unused	Returns204	Pass
TC-ADM-13	Upload Maintenance Images	With Png	Returns201 And Url	Pass
TC-ADM-14	Upload Maintenance Images	Without Files	Returns400	Pass
TC-ADM-15	Delete Category	When Category Missing	Returns404	Pass
TC-ADM-16	Get Users	With Role Filter	Lists Only That Role	Pass
TC-ADM-17	Generate Invoice	Twice For Same Order	Returns422	Pass
TC-ADM-18	Update Design Status	To Approved	Updates Status	Pass
TC-ADM-19	Update Design Status	With Unknown Status	Returns400	Pass
TC-ADM-20	Admin Endpoints	With Customer Token (case 1)	Return403	Pass
TC-ADM-21	Admin Endpoints	With Customer Token (case 2)	Return403	Pass
TC-ADM-22	Admin Endpoints	With Customer Token (case 3)	Return403	Pass
TC-ADM-23	Admin Endpoints	With Customer Token (case 4)	Return403	Pass
TC-ADM-24	Admin Endpoints	With Customer Token (case 5)	Return403	Pass
TC-ADM-25	Admin Endpoints	With Customer Token (case 6)	Return403	Pass
TC-ADM-26	Admin Endpoints	With Customer Token (case 7)	Return403	Pass
TC-ADM-27	Admin Endpoints	With Customer Token (case 8)	Return403	Pass
TC-ADM-28	Admin Endpoints	With Customer Token (case 9)	Return403	Pass
TC-ADM-29	Admin Endpoints	With Customer Token (case 10)	Return403	Pass
TC-ADM-30	Assign Maintenance	To Unknown Technician	Returns404	Pass
TC-ADM-31	Update Order Status	Pending Cash Order To Processing	Returns Updated Order	Pass
TC-ADM-32	Admin Endpoint	With Technician Token	Returns403	Pass
TC-ADM-33	Update Appointment Status	When Appointment Missing	Returns404	Pass
TC-ADM-34	Reschedule Appointment	To Past Date	Returns400	Pass
TC-ADM-35	Get Users	Without Filter	Lists All Users	Pass
TC-ADM-36	Deactivate User	When User Missing	Returns404	Pass
TC-ADM-37	Set Feedback Visibility	Without Hidden Param	Returns400	Pass
TC-ADM-38	Create Category	With Existing Name	Returns422	Pass
TC-ADM-39	Set Feedback Visibility	Hidden	Hides Review	Pass
TC-ADM-40	Create Category	With Blank Name	Returns400	Pass
TC-ADM-41	Generate Invoice	For Cancelled Order	Returns422	Pass
TC-ADM-42	Get All Designs	—	Lists Designs	Pass

ID	Operation	Condition / input	Expected output	Status
TC-ADM-43	Get All Feedback	—	Lists Reviews	Pass
TC-ADM-44	Update Appointment Status	Back To Pending	Returns422	Pass
TC-ADM-45	Update Order Status	Shipped Cash Order To Delivered	Marks Paid	Pass
TC-ADM-46	Update Category	When Category Missing	Returns404	Pass
TC-ADM-47	Update Design Status	When Design Missing	Returns404	Pass
TC-ADM-48	Assign Maintenance	With Past Deadline And No Cost	Returns400	Pass
TC-ADM-49	Create User	With Valid Technician	Returns201	Pass
TC-ADM-50	Update Appointment Status	Approve Pending	Sets Meeting Link	Pass
TC-ADM-51	Deactivate User	For Own Account	Returns422	Pass
TC-ADM-52	Admin Endpoint	Without Token	Is Denied	Pass
TC-ADM-53	Update Complaint Status	When Complaint Missing	Returns404	Pass
TC-ADM-54	Update Order Status	From Delivered	Returns409	Pass
TC-ADM-55	Get All Appointments	With Status Filter	Lists Only That Status	Pass
TC-ADM-56	Reschedule Appointment	When Completed	Returns422	Pass
TC-ADM-57	Create User	With Unknown Role And Short Password	Returns400	Pass
TC-ADM-58	Reschedule Appointment	To Future Date	Sets Rescheduled	Pass
TC-ADM-59	Update Complaint Status	To Resolved	Updates Status	Pass
TC-ADM-60	Get All Orders	With Status Filter	Lists Only That Status	Pass
TC-ADM-61	Activate User	For Inactive User	Marks Active	Pass
TC-ADM-62	Get All Orders	With Unknown Status	Returns400	Pass
TC-ADM-63	Assign Maintenance	To Customer	Returns422	Pass
TC-ADM-64	Upload Maintenance Images	With Disallowed Type	Returns400	Pass
TC-ADM-65	Get Dashboard	With Orders And Low Stock	Returns Stats	Pass
TC-ADM-66	Generate Invoice	For Delivered Order	Returns201 And Lists It	Pass
TC-ADM-67	Update Appointment Status	With Non Http Meeting Link	Returns422	Pass
TC-ADM-68	Set Feedback Visibility	When Review Missing	Returns404	Pass
TC-ADM-69	Update Order Status	When Order Missing	Returns404	Pass
TC-ADM-70	Create User	With Existing Email	Returns409	Pass
TC-ADM-71	Generate Invoice	When Order Missing	Returns404	Pass
TC-ADM-72	Update Category	With New Name	Returns Renamed Category	Pass
TC-ADM-73	Get All Complaints	—	Lists Complaints	Pass

CHAPTER 4 BACKEND UNIT TEST CASES

4.1 AppointmentServiceImplTest – FR023, FR024

ID	Unit under test	Condition / input	Expected result	Status
UT-APPOIN-01	Cancel Appointment	—	Should Throw When Already Completed	Pass
UT-APPOIN-02	Admin Reschedule	—	Should Store Note	Pass
UT-APPOIN-03	Admin Reschedule	—	Should Throw When Not Found	Pass
UT-APPOIN-04	Update Status	—	Pending Should Be Rejected	Pass
UT-APPOIN-05	Cancel Appointment	—	Should Allow Cancelling Approved Appointment	Pass
UT-APPOIN-06	Reschedule	—	Should Throw When Appointment Is Completed	Pass
UT-APPOIN-07	Admin Cancel	—	Should Throw When Completed	Pass
UT-APPOIN-08	Approve Appointment	—	Should Throw When Already Approved	Pass
UT-APPOIN-09	Book Appointment	—	Should Save Product Line And Mention It To Admins	Pass
UT-APPOIN-10	Reschedule	—	Should Update Date For Pending Appointment	Pass
UT-APPOIN-11	Approve Appointment	—	Should Reject Non Http Link	Pass
UT-APPOIN-12	Book Appointment	—	Should Store Uploaded Images	Pass
UT-APPOIN-13	Book Appointment	—	Should Notify Admins In App And By Email	Pass
UT-APPOIN-14	Admin Reschedule	—	Should Throw When Cancelled	Pass
UT-APPOIN-15	Approve Appointment	—	Should Set Status To Approved And Notify Customer	Pass
UT-APPOIN-16	Admin Cancel	—	Should Store Note And Include It In Notification	Pass
UT-APPOIN-17	Update Status	—	Approved Should Approve Pending Appointment	Pass
UT-APPOIN-18	Admin Reschedule	—	Should Work On Another Users Appointment	Pass
UT-APPOIN-19	Update Status	—	Rescheduled Should Be Rejected	Pass
UT-APPOIN-20	Update Status	—	Cancelled Should Cancel Appointment	Pass
UT-APPOIN-21	Book Appointment	—	Should Reject Outside Image Links	Pass
UT-APPOIN-22	Approve Appointment	—	Should Throw When Cancelled	Pass
UT-APPOIN-23	Cancel Appointment	—	Should Throw When Already Cancelled	Pass
UT-APPOIN-24	Approve Appointment	—	Should Store Meeting Link And Note And Send Them	Pass
UT-APPOIN-25	Completing	Future Appointment	Is Rejected	Pass
UT-APPOIN-26	Approve Appointment	—	Should Allow Rescheduled Appointment	Pass
UT-APPOIN-	Get Appointments By	—	Should Return User	Pass

ID	Unit under test	Condition / input	Expected result	Status
27	User		Appointments	
UT-APPOIN-28	Reschedule	—	Should Throw When Appointment Is Cancelled	Pass
UT-APPOIN-29	Book Appointment	—	Should Reject Unknown Product Line	Pass
UT-APPOIN-30	Cancel Appointment	—	Should Throw When Accessed By Different User	Pass
UT-APPOIN-31	Update Status	—	Completed Should Store Note	Pass
UT-APPOIN-32	Admin Cancel	—	Should Throw When Already Cancelled	Pass
UT-APPOIN-33	Get All Appointments	—	Should Return Every Appointment	Pass
UT-APPOIN-34	Approve Appointment	Without Link	Should Approve For In Person Meeting	Pass
UT-APPOIN-35	Admin Reschedule	—	Should Set New Date And Rescheduled Status	Pass
UT-APPOIN-36	Cancel Appointment	—	Should Set Status To Cancelled	Pass
UT-APPOIN-37	Customer Cancel	—	Should Notify Admins In App Only	Pass
UT-APPOIN-38	Approve Appointment	—	Should Reject Malformed Link	Pass
UT-APPOIN-39	Book Appointment	—	Should Save And Return Pending Appointment	Pass
UT-APPOIN-40	Approving	Appointment Whose Time Passed	Is Rejected	Pass
UT-APPOIN-41	Reschedule	—	Should Throw When Accessed By Different User	Pass
UT-APPOIN-42	Approve Appointment	—	Should Throw When Not Found	Pass
UT-APPOIN-43	Reschedule	—	Should Set Approved Appointment Back To Pending	Pass
UT-APPOIN-44	Book Appointment	—	Should Throw When User Not Found	Pass
UT-APPOIN-45	Get Appointments By Status	—	Should Return Matching Appointments	Pass
UT-APPOIN-46	Admin Cancel	—	Should Set Status To Cancelled And Notify Customer	Pass
UT-APPOIN-47	Update Status	—	Completed Should Throw When Not Approved	Pass
UT-APPOIN-48	Update Status	—	Completed Should Complete Approved Appointment	Pass

4.2 OrderServiceImplTest – FR012–FR018

ID	Unit under test	Condition / input	Expected result	Status
UT-ORDER-01	Payment With Wrong Amount	—	Is Refunded	Pass
UT-ORDER-02	Update Order Status	Shipped	Should Email Customer – but Processing Should Not	Pass
UT-ORDER-03	Update Order Status	Cancel	Should Restock And Cancel Unpaid Payment	Pass
UT-ORDER-04	Start Payment	—	Should Create New Payment When Previous Was	Pass

ID	Unit under test	Condition / input	Expected result	Status
			Cancelled	
UT-ORDER-05	Start Payment	First Time	Creates Payment For Exact Total	Pass
UT-ORDER-06	Update Order Status	—	Cannot Cancel After Shipping	Pass
UT-ORDER-07	Start Payment	—	Should Reuse Open Payment	Pass
UT-ORDER-08	Get Order By Id	—	Should Throw When Order Belongs To Different User	Pass
UT-ORDER-09	Place Order	—	Should Reject When Not Enough Stock	Pass
UT-ORDER-10	Confirm Payment	—	Should Stay Pending While Card Still Needed	Pass
UT-ORDER-11	Place Order	—	Should Throw When Cart Is Empty	Pass
UT-ORDER-12	Update Order Status	—	Should Succeed For Non Terminal Order	Pass
UT-ORDER-13	Start Payment	—	Should Report Already Paid And Record It	Pass
UT-ORDER-14	Get Order By Id	—	Should Throw When Order Not Found	Pass
UT-ORDER-15	Update Order Status	—	Failed Refund Alerts Admins	Pass
UT-ORDER-16	Update Order Status	—	Should Throw For Delivered Order	Pass
UT-ORDER-17	Get Orders By User	—	Should Return User Orders	Pass
UT-ORDER-18	Expire Unpaid Card Orders	—	Should Keep Order Paid Meanwhile	Pass
UT-ORDER-19	Update Order Status	Delivered Cash On Delivery	Should Mark Paid	Pass
UT-ORDER-20	Place Order	Card	Should Fail When Stripe Is Not Configured	Pass
UT-ORDER-21	Update Order Status	—	Should Block Processing Unpaid Card Order	Pass
UT-ORDER-22	Place Order	—	Should Throw When Cart Does Not Exist	Pass
UT-ORDER-23	Place Order	—	Rejects Archived Product	Pass
UT-ORDER-24	Confirm Payment	—	Should Reject Other Customers Order	Pass
UT-ORDER-25	Get All Orders	—	Should Return All Orders	Pass
UT-ORDER-26	Update Order Status	—	Same Status Is Rejected	Pass
UT-ORDER-27	Confirm Payment	—	Should Mark Paid When Stripe Says Succeeded	Pass
UT-ORDER-28	Start Payment	—	Should Reject Cash On Delivery Order	Pass
UT-ORDER-29	Update Order Status	Cancelling Paid Card Order	Refunds Through Stripe	Pass
UT-ORDER-30	Place Order	Cash On Delivery	Should Price Reserve Stock Clear Cart And Confirm	Pass
UT-ORDER-31	Update Order Status	—	Cannot Skip Steps	Pass
UT-ORDER-32	Get Orders By User	—	Should Throw When User Not Found	Pass
UT-ORDER-33	Update Order Status	—	Should Throw For Cancelled Order	Pass

ID	Unit under test	Condition / input	Expected result	Status
UT-ORDER-34	Get Checkout Summary	—	Should Use Server Pricing	Pass
UT-ORDER-35	Handle Stripe Webhook	—	Should Mark Matching Order Paid	Pass
UT-ORDER-36	Expire Unpaid Card Orders	—	Should Cancel And Restock	Pass
UT-ORDER-37	Payment Arriving For Cancelled Order	—	Is Refunded Not Kept	Pass
UT-ORDER-38	Place Order	Card	Should Reserve Stock Without Calling Stripe	Pass

4.3 ProductServiceImplTest – FR005–FR008

ID	Unit under test	Condition / input	Expected result	Status
UT-PRODUC-01	Delete Product	Not Found	Throws And Does Not Delete	Pass
UT-PRODUC-02	Update Product	Not Found	Throws	Pass
UT-PRODUC-03	Get All Products	—	Returns Mapped Responses	Pass
UT-PRODUC-04	Get Product By Id	Not Found	Throws	Pass
UT-PRODUC-05	Get Product By Id	Found	Returns Response	Pass
UT-PRODUC-06	Get All Products	—	Hides Archived Products	Pass
UT-PRODUC-07	Create Product	Category Not Found	Throws And Does Not Save	Pass
UT-PRODUC-08	Get Products By Category	Category Not Found	Throws	Pass
UT-PRODUC-09	Create Product	Cover Not In Gallery	Defaults To First Image	Pass
UT-PRODUC-10	Get Products By Material	—	Filters Case Insensitively	Pass
UT-PRODUC-11	Create Product	With Gallery	Keeps Chosen Cover	Pass
UT-PRODUC-12	Create Product	Null Stock	Defaults To Zero	Pass
UT-PRODUC-13	Upload Images	—	Delegates To File Storage In Products Folder	Pass
UT-PRODUC-14	Get Products By Category	Found	Returns Responses	Pass
UT-PRODUC-15	Delete Product	—	Archives Instead Of Deleting	Pass
UT-PRODUC-16	Update Product	Success	Updates Fields	Pass
UT-PRODUC-17	Create Product	Success	Saves And Returns	Pass
UT-PRODUC-18	Set Active	—	Restores Archived Product	Pass
UT-PRODUC-19	Search Products	—	Delegates To Repository	Pass

4.4 FeedbackServiceImplTest – FR029

ID	Unit under test	Condition / input	Expected result	Status
UT-FEEDBA-01	Submit Feedback	—	Should Save Review For Delivered Product	Pass
UT-FEEDBA-02	Update Feedback	Owner Can Edit	And It Is Marked Edited	Pass
UT-FEEDBA-03	Update Feedback	—	Unchanged Review Is Not Marked Edited	Pass
UT-FEEDBA-04	Submit Feedback	—	Should Reject Before Delivery	Pass
UT-FEEDBA-05	Set Hidden	—	Toggles Visibility	Pass
UT-FEEDBA-06	Get Feedback By User	—	Should Return User Feedback List	Pass
UT-FEEDBA-07	Submit Feedback	—	Should Throw When Product Already Reviewed	Pass
UT-FEEDBA-08	Update Feedback	—	Dropping To Low Rating Alerts Admins	Pass
UT-FEEDBA-09	Update Feedback	—	Other User Is Rejected	Pass
UT-FEEDBA-10	Get Product Reviews	No Reviews	Returns Zero Average	Pass
UT-FEEDBA-11	Submit Feedback	—	Low Rating Should Alert Admins	Pass
UT-FEEDBA-12	Submit Feedback	—	Should Throw When Order Belongs To Different User	Pass
UT-FEEDBA-13	Submit Feedback	—	Should Reject Product Not In Order	Pass
UT-FEEDBA-14	Submit Feedback	—	Should Throw When User Not Found	Pass
UT-FEEDBA-15	Short Name	—	Handles Single And Multiple Names	Pass
UT-FEEDBA-16	Admin View	—	Includes Product Name Image And Customer Email	Pass
UT-FEEDBA-17	Submit Feedback	—	Should Throw When Order Not Found	Pass
UT-FEEDBA-18	Get Product Reviews	—	Should Average And Shorten Names	Pass

4.5 UserServiceImplTest – FR003, FR004

ID	Unit under test	Condition / input	Expected result	Status
UT-USER-01	Create User	—	Should Throw On Duplicate Email	Pass
UT-USER-02	Create User	—	Should Reject Unknown Role Instead Of Creating It	Pass
UT-USER-03	Change Password	—	Should Update Password When Current Password Correct	Pass
UT-USER-04	Get Profile	—	Should Throw When User Not Found	Pass
UT-USER-05	Activate User	—	Should Set Active To True	Pass
UT-USER-06	Create User	—	Should Save User With Correct Role	Pass
UT-USER-07	Update Profile Image	—	Should Store In Avatars Folder And Save Url	Pass
UT-USER-08	Get All Users	—	Should Return All Users	Pass

ID	Unit under test	Condition / input	Expected result	Status
UT-USER-09	Get Profile	—	Should Return User Response When Found	Pass
UT-USER-10	Update Profile Image	—	Should Throw When Nothing Stored	Pass
UT-USER-11	Deactivate User	—	Should Refuse The Last Active Admin	Pass
UT-USER-12	Deactivate User	—	Should Set Active To False	Pass
UT-USER-13	Remove Profile Image	—	Should Clear Url	Pass
UT-USER-14	Update Profile	—	Should Update Name When Provided	Pass
UT-USER-15	Change Password	—	Should Throw When Current Password Is Incorrect	Pass
UT-USER-16	Get Users By Role	—	Should Return Only Matching Role	Pass

4.6 GlobalExceptionHandlerTest – NFR008

ID	Unit under test	Condition / input	Expected result	Status
UT-GLOBAL-01	Handle Security Breach	—	Should Return401 With Reuse Code	Pass
UT-GLOBAL-02	Handle Business Rule	—	Should Return422 Unprocessable Entity	Pass
UT-GLOBAL-03	Handle Bad Credentials	—	Should Return401	Pass
UT-GLOBAL-04	Handle Disabled	—	Should Return403 With Deactivated Message	Pass
UT-GLOBAL-05	Handle Duplicate Email	—	Should Return409 Conflict	Pass
UT-GLOBAL-06	Handle Not Found	—	Should Return404 Not Found	Pass
UT-GLOBAL-07	Handle Status Transition	—	Should Return409 Conflict	Pass
UT-GLOBAL-08	Handle Validation	—	Should Return400 With All Field Errors	Pass
UT-GLOBAL-09	Handle Max Upload Size	—	Should Return400 With Custom Message	Pass
UT-GLOBAL-10	Handle Invalid File	—	Should Return400 Bad Request	Pass
UT-GLOBAL-11	Handle Generic	—	Should Return500 Internal Server Error	Pass

4.7 MaintenanceServiceImplTest – FR020, FR021

ID	Unit under test	Condition / input	Expected result	Status
UT-MAINTE-01	Assign	—	Rejects Deadline In The Past	Pass
UT-MAINTE-02	Assign	Custom Design	Stores Title And Images	Pass
UT-MAINTE-03	Assign	With Product	Links Product	Pass
UT-MAINTE-04	Assign	To Non Technician	Is Rejected	Pass
UT-MAINTE-05	Update Status	Invalid Status	Is Rejected	Pass
UT-	Assign	Without Product Or Title	Throws	Pass

ID	Unit under test	Condition / input	Expected result	Status
MAINTE-06				
UT-MAINTE-07	Assign	—	Notifies Technician In App And By Email	Pass
UT-MAINTE-08	Assign	—	Stores Deadline And Includes It In Technician Notice	Pass
UT-MAINTE-09	Update Status	By Assigned Technician	Completes And Notifies Admins	Pass
UT-MAINTE-10	Get Maintenance By Technician	—	Maps Custom Design Without Product	Pass
UT-MAINTE-11	Update Status	By Another Technician	Is Rejected	Pass

4.8 CartServiceImplTest – FR012

ID	Unit under test	Condition / input	Expected result	Status
UT-CART-01	Get Cart	—	Should Throw When User Not Found And No Cart	Pass
UT-CART-02	Get Cart	—	Should Return Existing Cart	Pass
UT-CART-03	Add Item	—	Should Add New Item When Not Already In Cart	Pass
UT-CART-04	Get Cart	—	Should Create Cart When None Exists	Pass
UT-CART-05	Clear Cart	—	Should Call Delete By Cart	Pass
UT-CART-06	Remove Item	—	Should Throw When Item Belongs To Different User	Pass
UT-CART-07	Update Item Quantity	—	Should Throw When Item Belongs To Different Cart	Pass
UT-CART-08	Update Item Quantity	—	Should Delete Item When Quantity Is Zero Or Less	Pass
UT-CART-09	Add Item	—	Should Throw When Stock Insufficient	Pass
UT-CART-10	Add Item	—	Should Update Quantity When Item Already In Cart	Pass

4.9 PasswordResetServiceImplTest – FR003

ID	Unit under test	Condition / input	Expected result	Status
UT-PASSWO-01	Reset Password	Valid Token	Updates Password And Signs Out Sessions	Pass
UT-PASSWO-02	Reset Password	Expired Token	Throws	Pass
UT-PASSWO-03	Reset Password	Unknown Token	Throws	Pass
UT-PASSWO-04	Request Reset	Unknown Email	Does Nothing	Pass
UT-PASSWO-05	Request Reset	—	Link Has No Double Slash	Pass
UT-PASSWO-06	Reset Password	Used Token	Throws	Pass
UT-PASSWO-07	Request Reset	Known User	Stores Hashed Token And Emails Link	Pass
UT-PASSWO-08	Request Reset	Deactivated User	Does Nothing	Pass

4.10 ComplaintServiceImplTest – FR028, FR030

ID	Unit under test	Condition / input	Expected result	Status
UT-COMPLA-01	Get Complaints By User	—	Maps Results	Pass
UT-COMPLA-02	Submit Complaint	About Someone Elses Order	Is Rejected	Pass
UT-COMPLA-03	Update Complaint Status	Resolved	Notifies Customer In App And By Email	Pass
UT-COMPLA-04	Submit Complaint	Product Not In Order	Is Rejected	Pass
UT-COMPLA-05	Update Complaint Status	Invalid Status	Is Rejected	Pass
UT-COMPLA-06	Submit Complaint	About Product In Own Order	Saves And Notifies Admins By Email	Pass
UT-COMPLA-07	Submit Complaint	Without Order	Is Allowed	Pass

4.11 NotificationServiceImplTest – FR025, FR027

ID	Unit under test	Condition / input	Expected result	Status
UT-NOTIFI-01	Send Notification	—	Unknown User Is Ignored	Pass
UT-NOTIFI-02	Notify Admins	—	In App Only When No Email Subject	Pass
UT-NOTIFI-03	Mark Read	—	Is Scoped To The Owner	Pass
UT-NOTIFI-04	Send Notification	—	Truncates Long Messages To Fit The Column	Pass
UT-NOTIFI-05	Notify Admins	—	Emails Active Admins And Skips Deactivated Ones	Pass
UT-NOTIFI-06	Send Notification	—	Saves Message Type And Link	Pass

4.12 JwtTokenProviderTest – FR003, NFR011

ID	Unit under test	Condition / input	Expected result	Status
UT-JWTTOK-01	Extract Username	—	Should Return Email From Token	Pass
UT-JWTTOK-02	Validate Token	—	Should Return True For Valid Token	Pass
UT-JWTTOK-03	Generate Token	—	Should Return Non Null Token	Pass
UT-JWTTOK-04	Validate Token	—	Should Return False For Wrong User	Pass
UT-JWTTOK-05	Extract Expiration	—	Should Return Future Date	Pass

4.13 AuthServiceImplTest – FR001, FR003

ID	Unit under test	Condition / input	Expected result	Status
UT-AUTH-01	Login	—	Should Authenticate And Return Token With Cookie	Pass

ID	Unit under test	Condition / input	Expected result	Status
UT-AUTH-02	Logout	—	Should Invalidate Token And Clear Cookie	Pass
UT-AUTH-03	Register	—	Should Create Customer Role When It Does Not Exist	Pass
UT-AUTH-04	Register	—	Should Throw When Email Already Exists	Pass
UT-AUTH-05	Register	—	Should Create User With Customer Role And Return Token	Pass

4.14 CheckoutPricingTest – FR012, FR016

ID	Unit under test	Condition / input	Expected result	Status
UT-CHECKO-01	Tax Rate Percent	—	Is Human Readable	Pass
UT-CHECKO-02	Calculate	—	Free Shipping From Threshold	Pass
UT-CHECKO-03	Calculate	—	Adds Flat Shipping And Vat On Items	Pass
UT-CHECKO-04	Calculate	—	Rounds Vat To Cents	Pass

4.15 FileStorageServiceImplTest – FR006

ID	Unit under test	Condition / input	Expected result	Status
UT-FILEST-01	Store Images	—	Saves Files And Returns Public Urls	Pass
UT-FILEST-02	Store Images	No Files	Throws	Pass
UT-FILEST-03	Store Images	—	Skips Empty Files	Pass
UT-FILEST-04	Store Images	Disallowed Extension	Throws	Pass

4.16 CustomUserDetailsTest – FR004

ID	Unit under test	Condition / input	Expected result	Status
UT-CUSTOM-01	Is Enabled	Deactivated User	Returns False	Pass
UT-CUSTOM-02	Is Enabled	Null Active	Treated As Active	Pass
UT-CUSTOM-03	Is Enabled	Active User	Returns True	Pass

4.17 CustomUserDetailsServiceTest – FR003

ID	Unit under test	Condition / input	Expected result	Status
UT-CUSTOM-01	Load User By Username	—	Should Return User Details When User Exists	Pass
UT-	Load User By Username	—	Should Throw Username Not	Pass

ID	Unit under test	Condition / input	Expected result	Status
CUSTOM-02			Found Exception When Missing	

4.18 ContactServiceImplTest – FR028

ID	Unit under test	Condition / input	Expected result	Status
UT-CONTAC-01	Send Message	—	Notifies Admins In App And By Email With Reply Address	Pass
UT-CONTAC-02	Send Message	Without Phone	Omits Phone Line	Pass

4.19 BackendApplicationTests – —

ID	Unit under test	Condition / input	Expected result	Status
UT-BACKEN-01	Context Loads	—	—	Pass

CHAPTER 5 FRONTEND TEST CASES

ID	Test file	Test case	Type	Status
FE-01	frontend/src/components/common/Pagination.test.jsx	Should list every page given 7 pages or fewer	Unit	Pass
FE-02	frontend/src/components/common/Pagination.test.jsx	Should show the first pages and a gap given the current page near the start	Unit	Pass
FE-03	frontend/src/components/common/Pagination.test.jsx	Should show gaps on both sides given the current page in the middle	Unit	Pass
FE-04	frontend/src/components/common/Pagination.test.jsx	Should show the last pages given the current page near the end	Unit	Pass
FE-05	frontend/src/components/common/Pagination.test.jsx	Should keep a page that is in range	Unit	Pass
FE-06	frontend/src/components/common/Pagination.test.jsx	Should pull the page back to the last page given fewer results	Unit	Pass
FE-07	frontend/src/components/common/Pagination.test.jsx	Should return page 1 given zero items or a page below 1	Unit	Pass
FE-08	frontend/src/components/common/Pagination.test.jsx	Should show the item range summary and mark the current page	Unit	Pass
FE-09	frontend/src/components/common/Pagination.test.jsx	Should show an empty message and disable navigation given no items	Unit	Pass
FE-10	frontend/src/components/common/Pagination.test.jsx	Should call onPageChange with the chosen page	Unit	Pass
FE-11	frontend/src/components/common/ScrollToTop.test.jsx	Should scroll to the top given the first page load	Unit	Pass
FE-12	frontend/src/components/common/ScrollToTop.test.jsx	Should scroll to the top again given navigation to another page	Unit	Pass
FE-13	frontend/src/utils/deadline.test.js	FormatDeadline should parse YYYY-MM-DD as a local date	Unit	Pass
FE-14	frontend/src/utils/deadline.test.js	TodayInputValue should return today as a zero-padded YYYY-MM-DD string	Unit	Pass
FE-15	frontend/src/utils/deadline.test.js	Should return null given no deadline	Unit	Pass
FE-16	frontend/src/utils/deadline.test.js	Should show a neutral label given a completed task, even if overdue	Unit	Pass
FE-17	frontend/src/utils/deadline.test.js	Should mark the task urgent given a past deadline	Unit	Pass
FE-18	frontend/src/utils/deadline.test.js	Should show "Due today" given today as the deadline	Unit	Pass
FE-19	frontend/src/utils/deadline.test.js	Should warn given a deadline within two days	Unit	Pass
FE-20	frontend/src/utils/deadline.test.js	Should show the plain date given a deadline three or more days away	Unit	Pass
FE-21	frontend/src/utils/formatters.test.js	Should format a number with two decimals given a numeric amount	Unit	Pass
FE-22	frontend/src/utils/formatters.test.js	Should accept numeric strings given an amount from the API	Unit	Pass
FE-23	frontend/src/utils/formatters.test.js	Should return LKR 0.00 given null or undefined	Unit	Pass
FE-24	frontend/src/utils/formatters.test.js	Should return a dash given an empty value	Unit	Pass
FE-25	frontend/src/utils/formatters.test.js	Should format an ISO date given a valid date string	Unit	Pass
FE-26	frontend/src/utils/formatters.test.js	Should map known statuses to their badge class	Unit	Pass
FE-27	frontend/src/utils/formatters.test.js	Should fall back to the pending	Unit	Pass

ID	Test file	Test case	Type	Status
		badge given an unknown status		
FE-28	frontend/src/utils/formatters.test.js	Should prefer the API message, then the API error, then the JS message	Unit	Pass
FE-29	frontend/src/utils/formatters.test.js	Should return a generic message given no usable detail	Unit	Pass
FE-30	frontend/src/utils/formatters.test.js	FormatOrderNo should prefix the id with #	Unit	Pass
FE-31	tests/frontend/src/api/axios.int.test.js	Should send the stored access token as a Bearer header	Integration	Pass
FE-32	tests/frontend/src/api/axios.int.test.js	Should refresh the token and retry once given a 401 on a protected endpoint	Integration	Pass
FE-33	tests/frontend/src/api/axios.int.test.js	Should refresh only once given several requests fail with 401 at the same time	Integration	Pass
FE-34	tests/frontend/src/api/axios.int.test.js	Should reject queued requests too given the shared refresh fails	Integration	Pass
FE-35	tests/frontend/src/api/axios.int.test.js	Should not try to refresh given a 401 from the login endpoint	Integration	Pass
FE-36	tests/frontend/src/api/axios.int.test.js	Should clear the session and announce logout given the refresh itself fails	Integration	Pass
FE-37	tests/frontend/src/api/axios.int.test.js	Should pass non-401 errors straight through	Integration	Pass